



RESEARCH ARTICLE

MISINFORMATION

Misinformation exploits outrage to spread online

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We tested a hypothesis that misinformation exploits outrage to spread online, examining generalizability across multiple platforms, time periods, and classifications of misinformation. Outrage is highly engaging and need not be accurate to achieve its communicative goals, making it an attractive signal to embed in misinformation. In eight studies that used US data from Facebook (1,063,298 links) and Twitter (44,529 tweets, 24,007 users) and two behavioral experiments (1475 participants), we show that (i) misinformation sources evoke more outrage than do trustworthy sources; (ii) outrage facilitates the sharing of misinformation at least as strongly as sharing of trustworthy news; and (iii) users are more willing to share outrage-evoking misinformation without reading it first. Consequently, outrage-evoking misinformation may be difficult to mitigate with interventions that assume users want to share accurate information.

Online sharing of misinformation—defined here as false and misleading information (1–4)—remains a major concern (5–8). Although estimates of the prevalence of misinformation vary widely (4, 9–11), it has been linked to increases in political polarization (12, 13), antidemocratic sentiment (14–16), and increased vaccine hesitancy (7, 17, 18). Yet despite investing in detecting and reducing misinformation, digital platforms have had only limited success in curbing its spread (19, 20).

We investigated the relationship between misinformation and moral outrage, a mixture of anger and disgust triggered by perceived moral transgressions (21–25). Moral outrage (henceforth “outrage”) has several distinctive properties that could promote the spread of misinformation. First, outrage is highly engaging: Social media posts that express outrage are liked and shared more, training users to express more outrage and ranking algorithms to amplify it (26, 27). Second, outrage expressions can serve communicative goals that do not depend on information accuracy, such as signaling loyalty to a political group or broadcasting a moral stance (12, 27–35). Consequently, outrage-evoking misinformation may be difficult to mitigate with interventions such as fact-checking or accuracy prompts that assume users want to share accurate information

(12, 34, 36). Third, individuals who express outrage are seen as more trustworthy (32). This suggests that news sources might gain a credibility advantage by posting outrageous content. Collectively, these features provide strong incentives for misinformation purveyors to generate outrage-evoking content.

We combined analyses of Twitter and Facebook data with behavioral experiments to test three key hypotheses about misinformation and outrage. First, does misinformation tend to evoke more outrage than that by trustworthy news? Past work is suggestive but suffers from several limitations. Misinformation triggers more emotion in general than does trustworthy news (16, 37–41), but it remains unclear whether this relationship holds for outrage, whose particular properties pose special challenges for developing effective countermeasures. Moreover, prior studies of emotion and misinformation are limited to single platforms and time periods, raising questions about generalizability (39, 40, 42). We addressed this limitation by analyzing data from Facebook and Twitter across multiple time periods (Table 1). Because defining and classifying misinformation remains controversial (1, 4, 11), generalizability is further limited for the majority of past studies that relied on a single classification strategy (1, 4). In this work, we achieved robustness by testing our hypothesis across multiple classification strategies. Last, because misinformation and trustworthy news tend to circulate in different networks, comparisons between them are often confounded by audience characteristics (43, 44). We addressed this limitation by analyzing outrage evoked by misinformation and trustworthy news shared within the same networks of users.

Second, we tested whether outrage facilitates the spread of misinformation. Although outrage increases sharing information in general (26), outrage might have a more limited

impact on misinformation sharing because it is reputationally costly (45). Because outrage expressions will tend to reach a larger-than-average audience, outrage might increase the reputational risks of sharing misinformation, making users more attentive to accuracy to mitigate these risks. Accordingly, emotional reactions to headlines increase discernment of misinformation from trustworthy news (42). Thus, the potential reputational costs of sharing outrage that turns out to be false or misleading might outweigh the potential benefits of outrage for spreading misinformation. Alternatively, the ability of outrage to signal trustworthiness (32) and group identity (28) might provide some insurance for the potential reputational costs of sharing misinformation. We addressed these possibilities by comparing the effects of outrage on sharing misinformation and trustworthy news.

Our third hypothesis investigated how outrage shapes psychological motives for sharing misinformation. Past work on information sharing distinguishes between “epistemic” motives (motives to share accurate information) and “nonepistemic” motives (any motives that are indifferent to accuracy, such as expressing group loyalty or habitually responding to a familiar stimulus) (12, 34, 36, 46, 47). We leveraged two kinds of data to examine how outrage affects these motives: discernment of true from false headlines in a behavioral experiment and sharing Facebook links without reading them first. Research on the reputational costs of sharing misinformation suggests that outrage could enhance epistemic motives (45). If so, outrage should increase discernment and reduce sharing without reading (because epistemically motivated users should want to evaluate the information in the links they share). Alternatively, research on political and intergroup communication suggests that outrage could amplify nonepistemic motives (12). If so, outrage should have no effect on discernment and increase sharing without reading.

Overview of studies

We report eight observational studies on Facebook and Twitter (total $N_{\text{Facebook}} = 1,063,298$ links, $N_{\text{Twitter}} = 44,529$ tweets, $N_{\text{Twitter}} = 24,007$ users) and two behavioral experiments in a simulated social media environment (total $N = 1475$). The Facebook and Twitter studies examined engagement with social media posts that contain web links that we classified as misinformation or trustworthy (Fig. 1 and Table 1). Following past work (2, 3, 10, 19, 35, 48–54), we classified links on the basis of the quality of their source (the parent web domain of the link shared in the post) (19) as assessed by professional organizations [supplementary materials (SM) materials and methods 2.1]. In this “source classification” approach, social media posts that share links originating in low- versus high-quality sources are classified as misinformation

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Table 1. Study overview. We curated parallel datasets from Facebook (studies 1a, 2a, 3a, and 4a) and Twitter (Studies 1b, 2b, 3b, and 4b), including data from 2017 (studies 1a and 1b) and 2020 to 2021 (Studies 2a, 2b, 3a, 3b, 4a, and 4b). We also conducted two behavioral studies (studies 5a and 5b). In our observational studies (1 to 4), we classified misinformation using three separate databases of news domains assessed for source quality. In our behavioral studies (5a and 5b), we used headlines fact-checked as true or false.

Study	Data source	Time period	News source	Nlinks/tweets	Nusers/participants
1a	*Facebook	January 2017 to July 2017	IRA articles from domains in Domain Dataset 1	9026 links	—
1b	Twitter			3329 tweets	1656 users
2a	*Facebook	August 2020 to February 2021	IRA domains in Domain Dataset 1	192,108 links	—
2b	Twitter			10,550 tweets	5236 users
3a	*Facebook	August 2020 to February 2021	Domain Dataset 2	211,535 links	—
3b	Twitter			16,617 tweets	7485 users
4a	*Facebook	August 2020 to February 2021	Domain Dataset 3	650,629 links	—
4b	Twitter			14,033 tweets	5848 users
5a	Prolific	January 2020 to December 2021	Snopes.com	—	730 participants
5b	Prolific			—	745 participants

*The URL Shares dataset does not provide data at the individual user level, and so the number of users is not available.

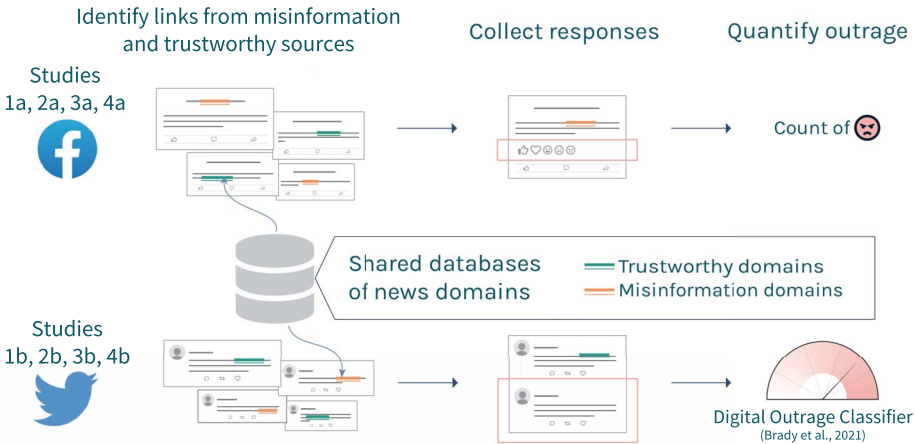


Fig. 1. Dataset curation. We identified links to misinformation and trustworthy sources using databases of parent web domains that had been assessed for news quality (SM materials and methods 2.1). We used the databases to curate pairs of datasets containing Facebook and Twitter posts that linked to the same articles or parent domains over identical time periods in 2016 and 2020. We then collected emotional responses to the links in each dataset. We quantified outrage on Facebook as a count of the Angry Reactions a link received and on Twitter as the proportion of responses that contained expressions of moral outrage as determined with our Digital Outrage Classifier (DOC) (SM materials and methods 5.1.2) (26). We refer to “Twitter” instead of “X” because our data were collected before the platform’s name was changed.

versus trustworthy information, respectively. This approach has practical and theoretical advantages over fact-checking individual articles, which is costly, prone to selection bias, difficult to scale, and focused on a tiny sliver of the broader misinformation ecosystem (19). Sources we classified as misinformation were more likely to produce content that was fact-checked as false or misleading, compared with sources we classified as trustworthy (supplementary text 5.1).

Our observational studies drew on three databases of parent web domains that used different criteria for classifying misinformation versus trustworthy news sources (Table 1 and SM materials and methods 2.1). We used each database to curate pairs of datasets containing

Facebook and Twitter posts that linked to the same articles (studies 1a and 1b) or parent domains (studies 2a and 2b, 3a and 3b, and 4a and 4b) over identical time periods in 2017 and 2020 to 2021. This approach enabled robustness tests across definitions of misinformation, platforms, and time periods. Studies 1 to 3 classified misinformation and trustworthy domains categorically, whereas study 4 assessed source quality on a continuous basis by using an existing dataset of news domains (55), which enabled a more fine-grained test of the relationship between source quality and outrage.

Studies 1a, 1b, 2a, and 2b contained links to misinformation and trustworthy articles and domains posted by the Internet Research

Agency (IRA), a Russian organization whose purpose was to sow disinformation and discord into American politics (SM materials and methods 3.1) (56, 57). These studies provided conservative hypothesis tests by comparing outrage responses to misinformation and trustworthy sources that presumably were all shared with provocative intent.

Studies 3a and 3b used an “audience matching” strategy to control for characteristics of networks that tend to circulate misinformation. We curated these datasets by identifying users who posted links from misinformation sources and subsequently identifying links from trustworthy sources posted by the same users. We then collected engagement data for Facebook and Twitter posts that link to misinformation and trustworthy sources shared by the same users (SM materials and methods 6.1.3).

To address the limitations of source classification (most notably the imperfect correspondence between source quality and misinformation at the individual article level), our behavioral experiments (studies 5a and 5b) examined responses to headlines individually fact-checked as true or false that we selected to evoke high or low outrage (SM materials and methods 7.1.2). These studies also enabled causal inferences about the effects of outrage on sharing and discernment of true from false headlines. In each study, American participants viewed 20 news headlines that varied on trustworthiness (true versus false) and outrage evocation (high versus low) and rated their likelihood of sharing it (study 5a) or its perceived accuracy (study 5b).

All Facebook studies drew on the URL Shares dataset, which is privacy-protected by using an implementation of differential privacy that involves the addition of pseudo-random noise to the data (56, 57). For such data, it is not possible to extract traditional estimates from our models, include interaction terms, or estimate

P values (56). Instead, we ran regressions that simulated the added noise across tens of thousands of models by sampling values from a noise distribution with a known variance, yielding average coefficient estimates across simulations as well as adjusted standard errors [we calculated confidence intervals (CIs) as ± 1 adjusted standard error above and below the coefficient estimate] (SM materials and methods 4.1.2) (58).

Results

Misinformation sources evoke more outrage than do trustworthy news sources

In studies 1a, 2a, 3a, and 4a (Facebook), we regressed the count of “Anger Reactions” for each link onto news source, in which news source was either dummy coded (misinformation versus trustworthy; studies 1a, 2a, and 3a) or continuous (low or high source quality; study 4a). Across all studies, links from misinformation sources were associated with more Anger Reactions than that of links from trustworthy sources [study 1a, regression coefficient (β) = 1.63, CIs = (1.58, 1.69); study 2a, β = 2.33, CIs = (2.31, 2.34); study 3a, β = 1.23, CIs = (1.21, 1.24); study 4, β = 1.97, CIs = (1.95, 1.98); mean β = 1.79] (Fig. 2). This association remained when controlling for audience size (supplementary text 1.2). Links from misinformation sources were more likely to evoke Anger Reactions than other emotions (“Love,” “Happy,” “Sad,” or “Wow”) (supplementary text 1.1).

In studies 1b, 2b, 3b, and 4b (Twitter), we regressed a binary variable denoting the presence or absence of outrage in tweet responses onto the news source linked in the original tweet (misinformation versus trustworthy in studies 1b, 2b, and 3b; low or high source quality in study 4b). Responses to links from misinformation sources were significantly more likely to contain outrage across all studies [study 1b, odds ratio (OR) = 2.66, P < 0.001, 95% CIs = (2.28, 3.12); study 2b, OR = 1.87, P < 0.001, 95% CIs = (1.72, 2.04); study 3b, OR = 1.57, P < 0.001,

95% CIs = (1.45, 1.71); study 4b, OR = 1.35, P = 0.001, 95% CIs = (1.13, 1.62); mean OR = 1.86] (Fig. 2B). Tabulated results are provided in supplementary text 1.3, and the results of alternative models are provided in supplementary text 1.4 and 1.5. Complementing the Facebook results, links from misinformation sources were more likely to evoke outrage than negative sentiment in general (supplementary text 2.5).

Outrage facilitates the spread of misinformation

In studies 1a, 2a, and 3a (Facebook), we regressed the count of shares for links on the number of Anger Reactions that they received. The implementation of differential privacy in the URL Shares dataset prevented us from estimating interactions (SM materials and methods 4.1.2) (59), so we ran separate models for links to trustworthy and misinformation sources. Anger Reactions were associated with increased shares for trustworthy sources [study 1a, β = 0.77, CIs = (0.77, 0.78); study 2a, β = 0.40, CIs = (0.40, 0.40); and study 3a, β = 0.40, CIs = (0.40, 0.40); mean β = 0.52] and for misinformation sources [study 1a, β = 0.87, CIs = (0.87, 0.87); study 2a, β = 0.46, CIs = (0.46, 0.46); and study 3a, β = 0.44, CIs = (0.44, 0.44); mean β = 0.59]. The relationship between anger and sharing was robust to the inclusion of audience size as a covariate (supplementary text 1.2). The relationship was larger for misinformation as compared with trustworthy sources in all studies (tabulated models are available in supplementary text 2.1).

In studies 1b, 2b, and 3b (Twitter), we estimated models that predicted the count of shares the original tweets received as a function of the presence of outrage in responses to the tweets, news source, and their interaction. We found a main effect of outrage on sharing in each Twitter study [study 1b, OR = 2.42, P < 0.001, 95% CIs = (1.91, 3.09); study 2b, OR = 5.39, P < 0.001, 95% CIs = (4.91, 5.92); and study 3b, OR = 10.15, P < 0.001, 95% CIs =

(9.43, 10.93); mean OR = 5.99]. This relationship was confirmed for trustworthy sources [study 1b, OR = 2.42, P < 0.001; study 2b, OR = 5.39, P < 0.001; and study 3b, OR = 10.15, P < 0.001] and misinformation sources [study 1b, OR = 3.31, P < 0.001; study 2b, OR = 9.05, P < 0.001; and study 3b, OR = 6.89, P < 0.001]. The interaction between outrage and news type was inconsistent across studies. In studies 1b and 2b, the effect of outrage on shares was stronger for misinformation than trustworthy news sources. In study 3b, the effect was stronger for trustworthy news sources as compared with misinformation (supplementary text 2.1). This pattern of results was robust to the inclusion of negative sentiment as a covariate (supplementary text 2.5).

Study 5a tested whether outrage evocation causally increased sharing intentions for misinformation and trustworthy news (Fig. 3). Participants were more likely to share high-outrage-evoking headlines compared with low-outrage-evoking headlines [β = 0.25, P = 0.003, 95% CIs = (0.09, 0.40)] and equally likely to share misinformation and trustworthy news [β = -0.08, P = 0.29, 95% CIs = (-0.24, 0.07)]. We found no interaction between outrage and news type [β = -0.002, P = 0.09, 95% CIs = (-0.31, 0.31)], suggesting that outrage-evoking headlines are shared more, regardless of whether they are trustworthy or misinformation. These results were robust to controlling for participants' political ideology (supplementary text 2.7) and were replicated in a hierarchical logistic model that regressed binarized willingness to share ratings (likely versus unlikely to share) on outrage evocation, news type, and their interaction (supplementary text 2.6).

Outrage increases nonepistemic motives for sharing

Next, we investigated the effects of outrage on motives for sharing. We first examined our Facebook data (studies 1a, 2a, and 3a) to test whether outrage-evoking links were shared

Fig. 2. Links to lower-quality news sources evoke more outrage.

(A) Study 4a results. On Facebook, links with lower source quality were associated with higher counts of Anger Reactions. None of the other emotion reactions were as strongly associated with source quality. **(B)** Study 4b results. On Twitter, links to domains with lower source quality had a higher probability of evoking outrage in responses. The relationship between source quality and negative sentiment was nonsignificant. Shaded areas indicate 95%CIs from negative binomial models. n.s., not significant; *** P = 0.001.



more without being read first, compared with links that evoked relatively less outrage. Because it is difficult to assess the accuracy of an article without reading it first, we took sharing-without-reading as an imperfect but informative proxy for the relative strength of nonepistemic (versus epistemic) motives. We regressed the count of sharing-without-reading on the count of Anger Reactions to links from misinformation and trustworthy sources. We found that Anger Reactions were a positive predictor of sharing-without-reading for links from misinformation sources [study 1a, $\beta = 0.63$, 95% CIs = (0.63, 0.63); study 2a, $\beta = 0.33$, 95% CIs = (0.33, 0.33); and study 3a, $\beta = 0.31$, 95% CIs = (0.31, 0.31)] as well as trust-

worthy sources [study 1a, $\beta = 0.50$, 95% CIs = (0.50, 0.50); study 2a, $\beta = 0.31$, 95% CIs = (0.31, 0.31); and study 3a, $\beta = 0.30$, 95% CIs = (0.30, 0.30)]. Across all studies, Anger Reactions more strongly predicted sharing-without-reading for misinformation than trustworthy sources (Fig. 4 and supplementary text 3.1). These results suggest that outrage increases the relative strength of nonepistemic (versus epistemic) motives for sharing. However, we observed similar effects for all emotional reactions, suggesting that emotions in general (beyond outrage in particular) affect nonepistemic motives for sharing (supplementary text 3.2).

As an additional test of how outrage affects motives for sharing, we turned to our behav-

ioral experiment (study 5b) that assessed the effects of outrage on discerning false from true headlines. Because epistemically motivated sharing depends on assessing information accuracy, we took discernment as an imperfect but informative proxy for epistemic motives. Participants accurately discerned false from true headlines: Trustworthy news was rated as more accurate than misinformation [$\beta = 0.65$, $P < 0.001$, 95% CIs = (0.39, 0.91)]. However, outrage did not significantly affect discernment [$\beta = 0.15$, $P = 0.59$, 95% CIs = (-0.38, 0.68)] (SM materials and methods 8.1). Thus, we did not find evidence that outrage influences epistemic motives for sharing.

Discussion

Across eight studies and two experiments that spanned multiple platforms, time periods, and definitions of misinformation, our findings suggest that (i) misinformation sources evoke more outrage than do trustworthy news sources; (ii) outrage facilitates the spread of misinformation at least as strongly as trustworthy news; and (iii) outrage enhances nonepistemic motives for sharing misinformation.

Our results suggest that outrage-evoking misinformation may be difficult to mitigate with countermeasures that focus on increasing epistemic motives, such as reminders to consider accuracy before sharing content (47, 60). Instead, our results are consistent with recent evidence that social media users sometimes share information that they know is inaccurate to satisfy nonepistemic motives such as signaling their political affiliation or moral stance (12), despite potential reputational costs (45, 61). We speculate that outrage-evoking misinformation may be less reputationally costly to share than other types of misinformation because of the signaling properties of outrage. If caught sharing misinformation, users can claim that they merely intended to express that the content is “outrageous if true” (62), preserving epistemic trust while bolstering their moral trust. Future studies might test this possibility directly, with an eye toward developing interventions that target nonepistemic (rather than epistemic) motives for sharing.

Our studies had several limitations. Many factors contribute to the spread of misinformation (43), whereas our work focused on moral outrage. We focused on US samples on Facebook and Twitter, and therefore our results might not generalize beyond this cultural setting or other social media platforms, such as Reddit or TikTok. Our observational studies followed prior work (3, 49, 50, 53) in classifying misinformation by using professionally assessed source quality ratings rather than relying on fact-checked classifications of articles as “true” or “false.” A major limitation of this source classification approach is that it requires inferring article quality from source quality,

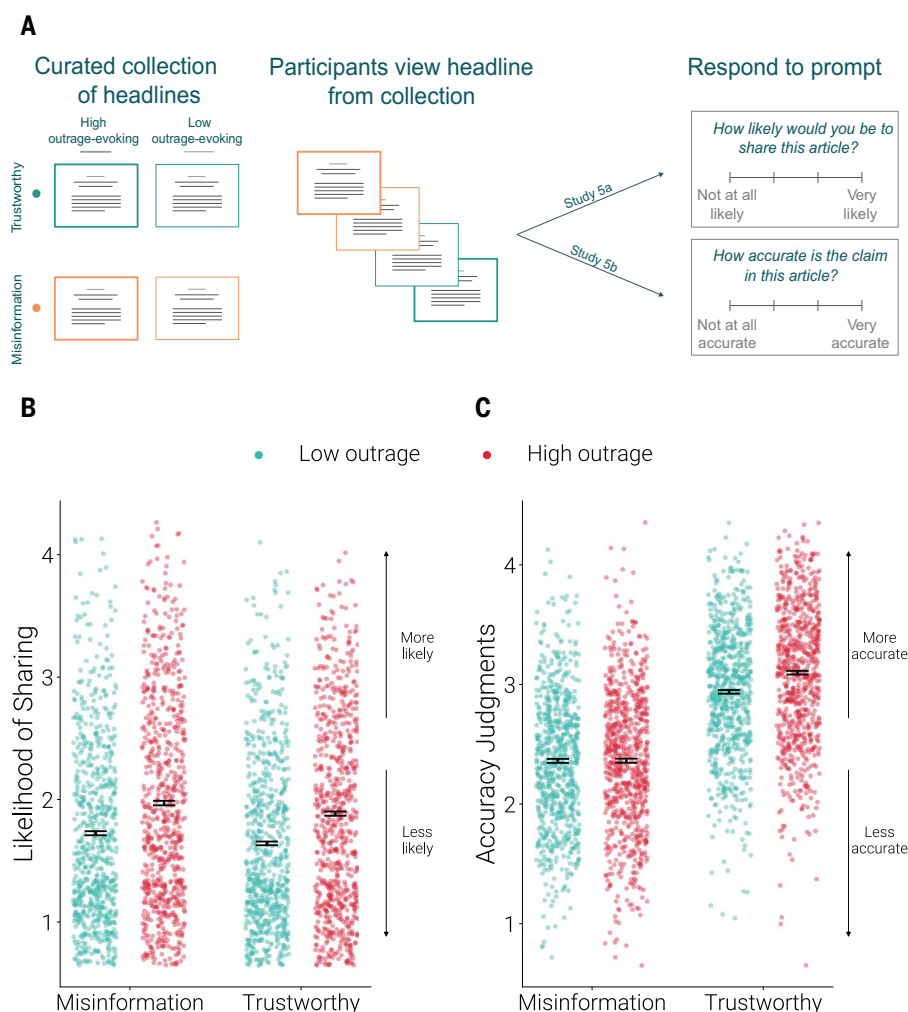


Fig. 3. Higher outrage increases sharing but does not affect accuracy discernment. (A) Design of studies 5a and 5b. Participants read a series of news headlines that were fact-checked as true or false. The headlines had been pilot tested so that half of those that participants read were outrage-evoking and the other half were not. After reading each headline, participants were asked how likely they would be to share it (study 5a) and how accurate they thought it was (study 5b). (B) Results of study 5a. Share ratings for high- and low-outrage-evoking headlines are shaded by news type (misinformation versus trustworthy). Dots indicate mean willingness to share, and error bars indicate SEM. High outrage evocation led to higher willingness to share ratings across misinformation and trustworthy news. (C) Results of study 5b. Accuracy judgments for high- and low-outrage-evoking headlines are shaded by news type (misinformation versus trustworthy). Dots indicate mean accuracy judgments, and error bars indicate SEM. The effect of outrage evocation on accuracy was nonsignificant for both misinformation and trustworthy news.

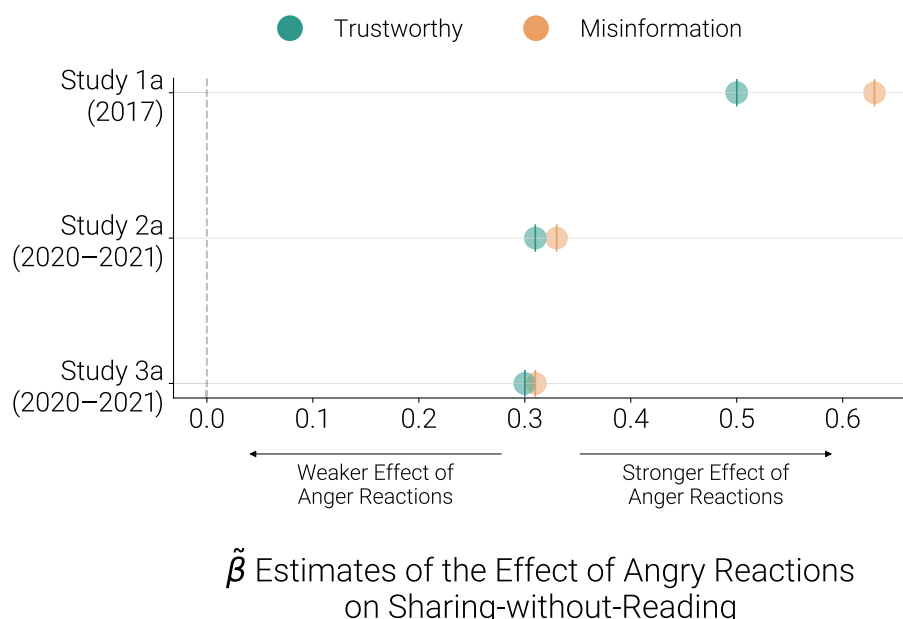


Fig. 4. Anger Reactions predict more sharing-without-reading for misinformation than for trustworthy sources. Effect size estimates from models regressing the count of sharing-without-reading on the count of Anger Reactions for misinformation (orange) and trustworthy (green) links in studies 1a, 2a, and 3a (Facebook). Error bars indicate regression $\pm$ 1 SE around estimates from differentially private regressions.

which may not always be valid. Constraints on Facebook data required that we operationalize outrage as a count of Anger Reactions; future work would benefit from a more specific measure of outrage, as with our Twitter analyses. Last, future work should explore alternative proxies for measuring epistemic and nonepistemic motives for sharing to further clarify their roles in spreading misinformation.

We take nonepistemic motives to broadly include any motive for sharing information that is not concerned with information accuracy. Recent work highlights habitual processes as a potential nonepistemic motive for sharing misinformation (26, 46). The observed association between outrage and sharing-without-reading is consistent with the possibility that outrage promotes habitual sharing that could inadvertently spread misinformation. This could arise because expressing outrage garners social rewards (26) that are delivered unpredictably, a schedule of reinforcement known to promote habit formation (63). Future work could examine this possibility more directly, as well as investigate the effects of outrage on other nonepistemic motives.

An outstanding challenge for a science of online behavior is how to observe and measure the influence of ranking algorithms on user behaviors, such as sharing misinformation (11, 64–66). Because outrage is associated with increased engagement online, outrage-evoking misinformation may be likely to spread farther in part because of the algorithmic amplification of engaging content. This is important because algorithms may up-rank news

articles associated with outrage, even if a user intended to express outrage toward the article for containing misinformation. Investigating this possibility is challenging, however, because of the opacity of platform algorithms and diminishing access to platform data (11).

Misinformation online remains a threat to a healthy digital public square and democracy (6, 14–16) and thus frequently is the subject of legal and policy directives that aim to mitigate these harms. Our findings suggest that misinformation exploits outrage to spread and offers concrete evidence for policy-makers to consider when attempting to craft effective solutions.

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and A.G. Methodology: K.L.M., W.J.B., A.G., B.K., and M.J.C. Project administration: K.L.M., W.J.B., and M.J.C. Supervision: W.J.B. and M.J.C. Visualization: K.L.M. Writing – original draft: K.L.M., W.J.B., and M.J.C. Writing – review and editing: W.J.B., A.G., B.K., K.K., and M.J.C. **Competing interests:** The authors declare that they have no competing interests. **Data and materials availability:** All data, code, and materials relating to studies 5a and 5b (behavioral studies) are available on OSF (67). Restrictions apply to the materials we can share from studies 1 to 4 (observational studies). Social Science One and Meta prohibit the sharing of data or analysis output related to the URL Shares dataset (59). Thus, for studies 1a, 2a, 3a, and 4a, we have only shared the analysis code. Researchers can apply for access to the URL Shares dataset here: <https://developers.facebook.com/docs/url-shares-dataset/overview>. X (formerly Twitter) also restricts how data from its platform can be shared (68). To comply with those restrictions, we have shared our data from in studies 1b, 2b, 3b,

and 4b without identifiable information about the tweets used. **License information:** Copyright © 2024 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. <https://www.science.org/about/science-licenses-journal-article-reuse>

SUPPLEMENTARY MATERIALS

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Materials and Methods
Supplementary Text
Figs. S1 and S2
Tables S1 to S34
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